

To,  
Editor-in-Chief, IEEE Control Systems Letters.

Dear Prof. Laura Menini,

We are pleased to submit the manuscript “Existence and Localization of a Limit Cycle in a Class of Benchmark Biomolecular Oscillators”, for your consideration of publication as a brief article.

Oscillatory behaviour is a fundamental biological phenomenon present in multiple contexts. However, the inherent nonlinearity and high dimensionality of mathematical models in biology makes proving the existence and the localization of limit cycle oscillations challenging. Here, we provided an elementary proof for the existence and a method for rigorously localizing the oscillatory solutions in a class of benchmark biomolecular oscillators. Specifically,

- We constructed a positive invariant set leveraging the boundedness in the dynamics and by removing a cylindrical hypervolume containing the unstable steady state.
- We demonstrated that a cross section of this set continuously maps to itself proving, by Brouwer’s Fixed Point theorem, the existence of a limit cycle.
- We employed an interval-based Reachability Analysis to localize the region of existence of the limit cycles within the cross-sections that map onto themselves.

We submit a substantially revised version of our previous submission (Paper ID: 25-0962). We believe that the combination of the Interval analytical methods with Brouwer’s Fixed Point theorem significantly adds to the novelty aspect. The rigorous localization can make the region of existence of limit cycle less conservative and, in principle, enclose the limit cycle as tightly as desired.

The resulting manuscript offers a unified, rigorous, and constructive approach that combines fixed-point theory with interval-based Reachability Analysis to establish the existence and localization of limit cycles in nonlinear biomolecular systems—an important and challenging problem in systems and synthetic biology. We believe this revised version provides improved clarity, strength, and applicability, making it well-suited for IEEE Control Systems Letters and of interest to the broader systems and control community.

Yours Sincerely,

Sidhanta Mohanty and Shaunak Sen